| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1(a) | 10 07 | **1** | |
| 1(b) | 123 | **2** | **M1** for 10 30 – 8 27 soi<br>or 10 30 – 8 52 + 25 soi or 25 + 50 + 48 |
| 1(c) | 25.2, $25\frac{1}{5}$ | **2** | **M1** for figs 29.4 ÷ 70 [× 60] oe |
| 1(d) | \$142.1[0] cao | **4** | **M2** for [adults =] 56 ÷ 8 × 5<br>and [child =] 56 ÷ 8 × 3 or better<br>or **M1** for 56 ÷ (5 + 3) × $k$ where $k$ = 1, 3 or 5<br>**M1** for *their* 35 × 2.80 + *their* 21 × 2.80 × $\frac{3}{4}$ oe |
| 2(a)(i) | Triangle drawn at (2, – 1), (2, – 4), (3, – 4) | **2** | **B1** for two correct points<br>If 0 scored, **SC1** for reflection of triangle $T$ in $y = -x$ |
| 2(a)(ii) | Triangle drawn at (– 5, 6), (– 2, 5), (– 5, 5) | **2** | **B1** for translation by $\begin{pmatrix}-1\\k\end{pmatrix}$ or by $\begin{pmatrix}k\\3\end{pmatrix}$<br>If 0 scored **SC1** for triangle drawn at (–4.5, 3.5), (–4.5, 4.5) and (–1.5, 3.5) |
| 2(a)(iii) | Enlargement<br>[SF] – 1.5 oe<br>[centre] (0, 3) | **3** | **B1** for each |
| 2(b) | 28.8, $28\frac{8}{10}$, $28\frac{4}{5}$ | **2** | **M1** for $1.2^2$ oe |
| 3(a)(i)(a) | 187 or 186.7 to 186.8 or $186\frac{42}{53}$ | **1** | |
| 3(a)(i)(b) | 2 : 7 : 42 cao | **2** | **B1** for 106 : 371 : 2226 or any equivalent ratio<br>If 0 scored, **SC1** for 2 : 7 : 42 in the wrong order |
| 3(a)(ii) | 33.3 or 33.28 to 33.29 | **2** | **M1** for $\frac{2967-2226}{2226}$ [× 100] oe<br>or $\frac{2967}{2226}$ × 100 [– 100] oe |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 3(a)(iii) | 1706 cao nfww | **3** | **B2** for 1705 to 1706.0… or 1710<br>or **M1** for $\left(1+\frac{30.48}{100}\right)x = 2226$ oe or better<br>If 0 or M1 scored, **SC1** for rounding *their* decimal answer seen to nearest integer |
| 3(b) | 3897 | **5** | **B1** for $a = 2000$<br>**M2** for [$b$ =] $\sqrt[3]{\frac{2662}{2000}}$<br>or **M1** for $2662 = 2000b^3$<br>**M1** for *their* $2000 \times \left(\sqrt[3]{\frac{2662}{their\,2000}}\right)^7$<br>or for *their* $a \times$ (*their* $b$)$^7$ provided *their* $a$ and *their* $b$ are clearly identified in the working<br>If 0 or M1 scored, **SC1** for rounding *their* decimal answer seen to nearest integer. |
| 4(a) | $\frac{(12-2)\times 180}{12}$ [= 150] oe<br>or $180 - \frac{360}{12}$ [= 150] | **1** | Accept $\frac{(2\times 12-4)\times 90}{12}$ [= 150] |
| 4(b)(i) | $\frac{3}{\cos 75}$ oe<br>or<br>$\frac{6\sin 75}{\sin 30}$ | **M2** | **M1** for $\frac{3}{AO} = \cos 75$ oe<br>or<br>$\frac{r}{\sin 75} = \frac{6}{\sin 30}$ |
| | 11.59… | **A1** | |
| 4(b)(ii)(a) | 72.8 or 72.9 or 72.82 to 72.89… | **2** | **M1** for $2 \times \pi \times 11.6$ |
| 4(b)(ii)(b) | 12.1 or 12.06 to 12.08 | **2** | **M1** for [6 +] *their* **(b)(ii)(a)** ÷ 12 oe |
| 4(c) | 806 or 807 or 805.9 to 807.4 | **3** | **B2** for 402.9… to 403.7<br>OR<br>**M2** for $\frac{1}{2}\times 6\times 11.6\times \sin 75\times 12\times 2$ oe<br>or **M1** for $\frac{1}{2}\times 6\times 11.6\times \sin 75\,[\times k]$ oe |
| 5(a)(i) | $20 < t \leqslant 35$ | **1** | |